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A Large Language Model, Instruction Data and Benchmark for Finance

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FIT: Financial Instruction Tuning Dataset

The FIRST instruction-tuning dataset with 128K instructions for finance with financial prediction and NLP tasks

Raw Data

15 Public Datasets on 8 Tasks

Data	Task	Raw	Instruction	Data Types	Modalities	License
FPB	sentiment analysis	4,845	48,450	news	text	CC BY-SA 3.0
FiQA-SA	sentiment analysis	1,173	11,730	news headlines,tweets	text	Public
Headlines	news headline classification	11,412	11,412	news headlines	text	CC BY-SA 3.0
FOMC	hawkish-dovish classification	496	496	FOMC transcripts	text	CC BY-NC 4.0
NER	named entity recognition	609	6,090	financial agreements	text	CC BY-SA 3.0
FiNER-ORD	named entity recognition	1,080	1,080	news articles	text	CC BY-SA 3.0
FinQA	question answering	8,281	8,281	earnings reports	text,table	MIT License
ConvFinQA	question answering	3,458	3,458	earnings reports	text,table	MIT License
ECTSum	text summarization	495	495	earning call transcipts	text	Public
EDTSum	text summarization	2,000	2,000	news articles	text	Public
BigData22	stock movement prediction	7,168	7,168	tweets,historical prices	text,time series	Public
ACL18	stock movement prediction	27,080	27,080	tweets,historical prices	text,time series	MIT License
CIKM18	stock movement prediction	4,971	4,971	tweets,historical prices	text,time series	Public
German	credit scoring	1,000	1,000	credit records	table	CC BY 4.0
Australia	credit scoring	690	690	credit records	table	CC BY 4.0

Instruction Construction

Tasks without Context:

Instruction: [task prompt] Text: [input text] Response: [output]

Tasks with Context:

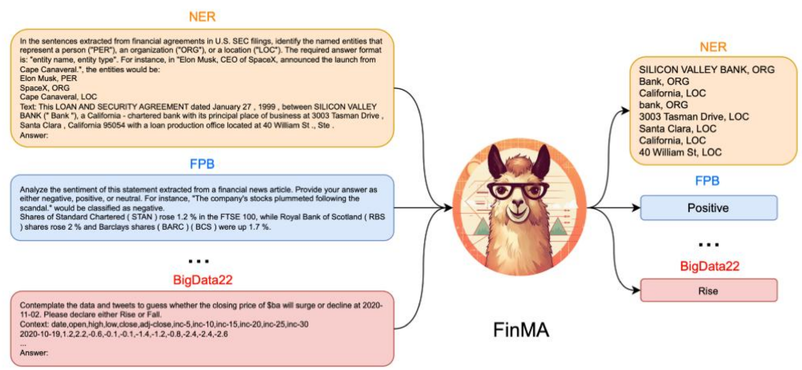
Instruction: [task prompt] Context: [input context] Question: [input question] Response: [answer]

Expert Hand-Crafted Instructions:

Data	Prompt
FPB	"Analyze the sentiment of this statement extracted from a financial news article. Provide your answer as either negative, positive or neutral. For instance, 'The company's stocks plummeted following the scandal.' would be classified as negative."
FiQA-SA	"What is the sentiment of the following financial {category}: Positive, Negative, or Neutral?"
Headlines	"Consider whether the headline mentions the price of gold. Is there a Price or Not in the gold commodity market indicated in the news headline? Please answer Yes or No."
NER	"In the sentences extracted from financial agreements in U.S. SEC filings, identify the named entities that represent a person ('PER'), an organization ('ORG'), or a location ('LOC'). The required answer format is: 'entity name, entity type'. For instance, in 'Elon Musk, CEO of SpaceX, announced the launch from Cape Canaveral.', the entities would be: 'Elon Musk, PER; SpaceX, ORG; Cape Canaveral, LOC'"
FiNER-ORD	"In the list of tokens, identify 'Person', 'Location', and 'Organisation' and label each accordingly. If the entity spans multiple tokens, use the prefix B-PER, B-LOC, or B-ORG for the first token, and I-PER, I-LOC, or I-ORG for the subsequent tokens of that entity. The beginning of each separate entity should always be labeled with a B-PER, B-LOC, or B-ORG prefix. If the token does not fit into any of the three named categories, or is not a named entity, label it as 'O'. Each line should contain one token and its corresponding label, separated by a colon. Do not combine tokens on your own. The format for each line should be: 'token:label'. Text: And all because you failed to prepare ! Answer:"
FinQA	"Given the financial data and expert analysis, please answer this question:"
ConvFinQA	"In the context of this series of interconnected finance-related queries and the additional information provided by the pretext, table data, and post text from a company's financial filings, please provide a response to the final question. This may require extracting information from the context and performing mathematical calculations. Please take into account the information provided in the preceding questions and their answers when formulating your response:"
BigData22	" Contemplate the data and tweets to guess whether the closing price of {tid} will surge or decline at {point}. Please declare either Rise or Fall."
ECTSum	"Given the following article, please produce a list of 0 and 1, each separated by ' ' to indicate which sentences should be included in the final summary. The article's sentences have been split by ' '. Please mark each sentence with 1 if it should be included in the summary and 0 if it should not."
EDTSum	"You are given a text that consists of multiple sentences. Your task is to perform abstractive summarization on this text. Use your understanding of the content to express the main ideas and crucial details in a shorter, coherent, and natural sounding text."
German	"Assess the creditworthiness of a customer using the following table attributes for financial status. Respond with either 'good' or 'bad'. And the table attributes including 13 categorical attributes and 7 numerical attributes are as follows:"
FOMC	"Examine the excerpt from a central bank's release below. Classify it as HAWKISH if it advocates for a tightening of monetary policy, DOVISH if it suggests an easing of monetary policy, or NEUTRAL if the stance is unbiased. Your response should return only HAWKISH, DOVISH, or NEUTRAL."

FinMA: Financial Large Language Model

Multi-task fine-tuned financial LLMs (Based on LLaMA 7B & 30B)



FLARE: Financial Evaluation Benchmark

The FIRST financial LLMs benchmark with financial NLP and application tasks

Data	Task	Valid	Test	Evaluation
FPB (Malo et al., 2014)	sentiment analysis	775	970	F1, Accuracy
FiQA-SA (Maia et al., 2018)	sentiment analysis	188	235	F1
Headlines (Sinha and Khandait, 2021)	news headline classification	1,141	2,283	Avg F1
NER (Alvarado et al., 2015)	named entity recognition	103	980	Entity F1
FiNER-ORD (Shah et al., 2023b)	named entity recognition	-	1080	Entity F1
FinQA (Chen et al., 2021)	question answering	883	1,147	EM Accuracy
ConvFinQA (Chen et al., 2022)	question answering	2,210	1,490	EM Accuracy
BigData22 (Soun et al., 2022)	stock movement prediction	798	1,470	Accuracy, MCC
ACL18 (Xu and Cohen, 2018)	stock movement prediction	2,560	3,720	Accuracy, MCC
CIKM18 (Wu et al., 2018)	stock movement prediction	431	1,140	Accuracy, MCC
ECTSum Mukherjee et al. (2022)	text summarization	-	495	ROUGE, BERTScore, BARTScore
EDTSum Zhou et al. (202)	text summarization	-	2000	ROUGE, BERTScore, BARTScore
German Hofmann (1994)	credit scoring	-	1000	F1, MCC
Australian Quinlan (1987)	credit scoring	-	690	F1, MCC
FOMC Shah et al. (2023a)	hawkish-dovish classification	-	496	F1, Accuracy

BloombergGPT: The only large language model with 50B parameters pretrained with the financial texts.
GPT-4: A powerful instruction following LLMs with around 1T parameters proposed by OpenAI.
ChatGPT: An instruction following LLMs with 175B parameters from OpenAI.
Vicuna-7B: An instruction following LLMs by fine-tuning LLaMA-7B.

Main Results

Dataset	Metrics	Chat GPT	GPT 4	Bloomberg GPT	Vicuna 7B	FinMA 7B	FinMA 7B-trade	FinMA 7B-full	FinMA 30B
FPB	F1	0.78*	0.78*	0.51*	0.29	**0.94**	0.03	**0.94**	0.88
	Acc	0.78*	0.76*	-	0.26	**0.94**	0.12	**0.94**	0.87
FiQA-SA	F1	0.60	0.80	0.75*	0.32	0.85	0.16	0.82	**0.87**
Headlines	AvgF1	0.77*	0.86*	0.82*	0.60	**0.97**	0.28	**0.97**	**0.97**
NER	EntityF1	0.77*	**0.83***	0.61*	0.12	0.59	0.00	0.64	0.62
FINER-ORD	EntityF1	0.28	**0.77**	-	0.00	0.00	0.00	0.00	0.00
FinQA	EmAcc	0.58*	**0.63***	-	0.00	0.06	0.00	0.04	0.11
ConvFinQA	EmAcc	0.60*	**0.76***	0.43*	0.00	0.25	0.00	0.20	0.40
BigData22	Acc	0.53	**0.54**	-	0.44	0.45	0.45	0.51	0.47
	MCC	-0.025	0.03	-	-0.05	0.02	0.00	0.02	**0.04**
ACL18	Acc	0.50	**0.52**	-	0.50	0.49	0.49	0.51	0.49
	MCC	0.005	0.02	-	0.02	-0.01	**0.03**	0.03	0.00
CIKM18	Acc	0.55	**0.57**	-	0.44	0.43	0.43	0.50	0.43
	MCC	0.01	0.02	-	-0.03	-0.02	-0.003	**0.08**	-0.05
EDTSUM	Rouge-1	0.17	0.2	-	**0.22**	0.09	0.05	0.13	0.17
	Rouge-2	0.08	0.09	-	**0.10**	0.04	0.02	0.06	0.08
	Rouge-N	0.13	0.15	-	**0.17**	0.08	0.05	0.10	0.14
	BertScore	0.66	**0.67**	-	0.61	0.56	0.51	0.38	0.54
	BartScore	-3.64	**-3.62**	-	-4.13	-6.12	-6.91	-5.71	-5.24
ECTSUM	Rouge-1	0.00	0.00	-	0.00	0.00	0.00	0.00	0.00
	Rouge-2	0.00	0.00	-	0.00	0.00	0.00	0.00	0.00
	Rouge-N	0.00	0.00	-	0.00	0.00	0.00	0.00	0.00
	BertScore	0.00	0.00	-	0.00	0.00	0.00	0.00	0.00
	BartScore	-5.18	-5.18	-	-5.18	-5.18	-5.18	-5.18	-5.18
German	F1	0.20	**0.55**	-	0.52	0.17	0.52	0.17	0.53
	MCC	-0.10	-0.02	-	**0.00**	0.00	-0.07	0.00	-0.07
Australian	F1	0.41	**0.74**	-	0.26	0.41	0.26	0.41	0.46
	MCC	0.00	**0.47**	-	0.00	0.00	0.00	0.00	-0.01
FOMC	F1	0.64	**0.71**	-	0.19	0.49	0.10	0.49	0.43
	Acc	0.6	**0.69**	-	0.28	0.47	0.25	0.46	0.53

Stock Prediction Results

Method	BIGDATA22		ACL18		CIKM18	
	ACC	MCC	ACC	MCC	ACC	MCC
Logistic regression (LR)	0.53	0.02	0.52	0.04	0.53	-0.04
Random forest (RF)	0.47	-0.11	0.52	0.03	0.54	0.01
LSTM	0.51	0.01	0.53	0.06	0.53	0.02
Attention LSTM (ALSTM)	0.49	-0.03	0.52	0.04	0.53	-0.01
Adv-ALSTM	0.50	0.01	0.53	0.07	0.54	0.02
DTML	0.52	0.07	0.58	0.18	0.54	-0.00
XGBoost	0.52	-0.04	0.49	-0.02	**0.58**	0.07
XGBRefressor	0.46	-0.13	0.50	-0.01	0.53	-0.03
ALSTM-W	0.48	-0.01	0.53	0.08	0.54	0.03
ALSTM-D	0.49	0.01	0.53	0.07	0.50	-0.04
StockNet	0.53	-0.00	0.54	-0.03	0.52	-0.02
SLOT	**0.55**	**0.10**	**0.59**	**0.21**	0.56	**0.09**
FinMA-7B	0.48	0.04	0.50	0.00	0.56	-0.02
FinMA-30B	0.47	0.04	0.49	0.00	0.43	-0.05
FinMA-7B-full	0.49	0.01	0.56	0.10	0.53	-0.03

Conclusions

1. Fine-tuned LLMs can perform well on Financial NLP tasks
2. LLMs struggle in Question Answering and Financial Forecasting Tasks
3. Better understanding and generation ability with larger model size
4. All resources including benchmark, instruction dataset, and models are publicly available in GitHub